Ramp energy lab

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Compare gravitational energy change with measured kinetic energy.

SUCCESS CRITERIA - I can...

1. Measure height vertically and speed at a stated location.
2. Calculate an energy balance.
3. Discuss rotation, resistance and uncertainty.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “rotation” and “thermal energy” to explain your answer. Sentence starters are optional.

rotation: turning about an axis

thermal energy: internal energy associated with microscopic motion/interactions

energy budget: an account of all energy stored and transferred

01 Warm-up: bring back an earlier idea

An ideal system has total mechanical energy **15 J**. At a point where U=**3 J**, find K.

Word help: mechanical energy = total kinetic and potential energy in a model

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: mg Δh = $\Delta K_{translation}$ + $\Delta K_{rotation}$ + $\Delta E_{thermal}$. Closed energy accounting near Earth; Δh is the positive drop. Find whether the object rolls or slides; use the approved ramp setup.

Worked reasoning: A supplied ramp run loses **6 J** of gravitational potential energy but gains only **4 J** of translational kinetic energy. The **2 J** difference needs an account, such as rotation or friction.

Your turn: Draw an energy account with the **2 J** remainder explicitly labeled as not yet identified.

Sentence starter: In my work, I used … to show … . I labeled it … .

Initial **Transfers** **Final**

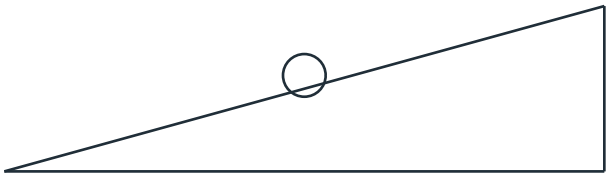
Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

On the ramp sketch, mark the vertical height h and the along-ramp path. Which belongs in mgh?

Sentence starter: In my work, I used … to show … . I labeled it … .



Label h vertically. The marker represents the object on the sloping surface.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

Supplied practice values: mass **1 kg** drops **0.50 m**; g=**10**. At bottom translational K=**4 J**.
1. **Find lost U** and energy not accounted for by translation alone.
2. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

05 Explain your reasoning

1. Create a table for repeated trials.
2. Include actual height, mass, bottom speed,ΔU, K and one observation about losses or rotation.

Sentence starter: I notice … . This supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Keep mass fixed and multiply speed by **4**. By what factor does kinetic energy change? Explain from the equation. **Show the energy changes** and transfers; **label units**.

Word help: kinetic energy = energy associated with motion

Sentence starter: I think … . I can point to …, which shows …, so … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

07 Exit check

A rolling object has less½mv² than its lost mgh.
1. **Name two possible destinations of** the difference.
2. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

1. Design a comparison that helps distinguish rolling energy from sliding friction as explanations for a lower translational speed.
2. **Use the energy** diagram to **show the quantities you compare**; **label units** and any unknowns.

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3